

SHAP-CAT: A INTERPRETABLE MULTI-MODAL FRAMEWORK ENHANCING WSI CLASSIFICATION VIA VIRTUAL STAINING AND SHAPLEY-VALUE-BASED MULTIMODAL FUSION

Jun Wang, Yu mao, Nan Guan, Chun Jason Xue,

{jwang699-c, yumao7-c}@my.cityu.edu.hk, nanguan@cityu.edu.hk jason.xue@mbzuai.ac.ae

ABSTRACT

The multimodal model has demonstrated promise in histopathology. However, most multimodal models are based on H&E and genomics, adopting increasingly complex yet black-box designs. In our paper, we propose a novel interpretable multimodal framework named SHAP-CAT, which uses a Shapley-value-based dimension reduction technique for effective multimodal fusion. Starting with two paired modalities – H&E and IHC images, we employ virtual staining techniques to enhance limited input data by generating a new clinical-related modality. Lightweight bag-level representations are extracted from image modalities and a Shapley-value-based mechanism is used for dimension reduction. For each dimension of the bag-level representation, attribution values are calculated to indicate how changes in the specific dimensions of the input affect the model output. In this way, we select a few top important dimensions of bag-level representation for each image modality to late fusion. Our experimental results demonstrate that the proposed SHAP-CAT framework incorporating synthetic modalities significantly enhances model performance, yielding a 5% increase in accuracy for the BCI, an 8% increase for IHC4BC-ER, and an 11% increase for the IHC4BC-PR dataset.

1 INTRODUCTION

Recent advances in artificial intelligence have significantly impacted histopathology, particularly through the development of multimodal models. These models integrate various types of data, such as whole slide images and molecular profiles, to improve diagnosis, prediction, and treatment personalization (Chen et al., 2022; Boehm et al., 2022; Chen et al., 2020). Recent efforts are expanding to include multi-staining images like IHC (Jaume et al., 2024; Wang et al., 2024; Foersch et al., 2023) and Trichrome-stained WSIs (Dwivedi et al., 2022) for better identification of specific molecular features related to cancer. The integration of diverse modalities is crucial, since different image modalities carry different information related to cancer (Perez-Lopez et al., 2024; Boehm et al., 2022; Stahlschmidt et al., 2022).

However, there are still many technical, analytical, and clinical challenges that are amplified in the presence of multimodal data.

- Limited public paired datasets (Steyaert et al., 2023; Miotto et al., 2018; Perez-Lopez et al., 2024): Developing multimodal models require modality-paired and datasets with labels. The data also needs to be complete and large in sample.
- Most multimodal models for histopathology are the combination of molecular features and WSIs, not different types of WSIs. Although molecular data are of great relevance to precision medicine, they don't have tissue structure, spatial, and morphological information (Alturkistani et al., 2016).
- Very complex and different multimodal fusion technique with *low interpretability*: Li et al. (2022); Wang et al. (2021); Lipkova et al. (2022) have complex design such as hierarchy

fusion, intermediate gradual fusion, and intermediate guided-fusion, but they ignore that medical imaging domain requires models to be interpretable.

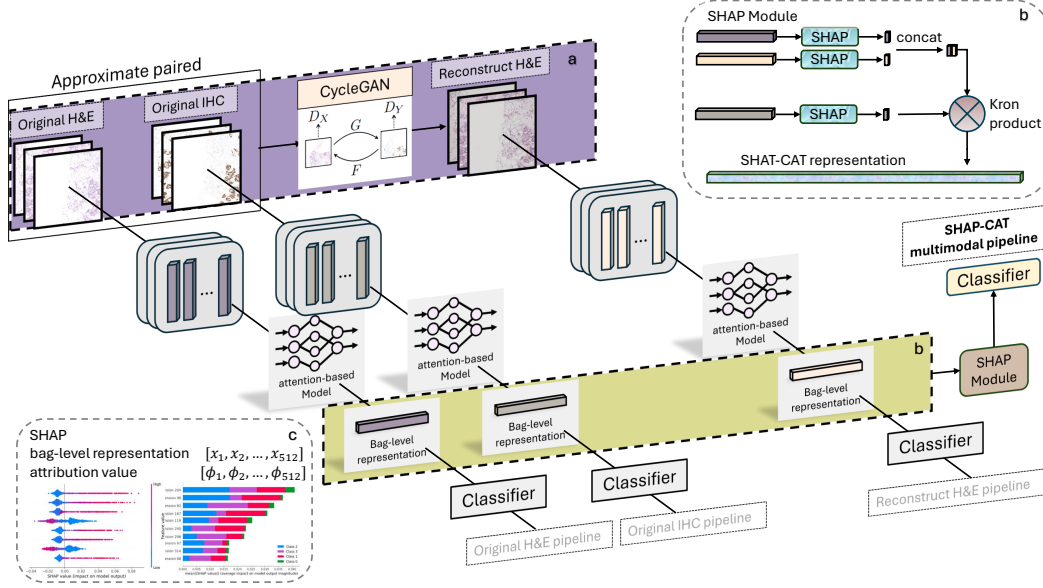


Figure 1: Our proposed **SHAP-CAT framework**, which includes three Parallel Feature Extraction Pipelines for different modalities and a SHAP-CAT pipeline for multimodal representation predictions. (a) Generating a new modality by a pre-trained CycleGAN. (b) Extract bag-level representations for each modality from the Parallel Feature Extraction Pipeline and adopt the SHAP pool to reduce dimensions for further late fusion. (c) The descriptions of our key idea of how to select the top important dimensions for reduction. The x-axis represents the attribution value, the y-axis ranks features by the magnitude of absolute attributions, and the color indicates the feature value. It’s important to note that the meaning of feature values are black-box and hard to interpret. By applying attribution values, the impact of features can be understood, and both positive attribution values and negative attribution values contribute to the output. (c) left shows the SHAP values of each dimension across all samples within a single class, while the right side shows the mean absolute value of the SHAP values for each dimension, broken down by class in multi-class tasks.

Given the difficulty of obtaining quality datasets (*the first challenge*), we propose a virtual staining-based multimodal framework that uses H&E, IHC and one more generated modality for WSI classifications. Our multimodal network is capable of integrating triple image modalities in weakly supervised learning on cancer grading tasks (*the second challenge*). After training the specific pipeline and extracting the bag-level representations for each modality, our framework uses the Shapley-value-based approach of dimension reduction for further multimodal fusion, avoiding the curse of dimensions and demonstrating high interpretability (*the third challenge*). For a given set of bag-level representations belonging to a patient sample, we employ a Shapley-value-based method to characterize the importance of each dimension within the feature space. This method attributes the predictions of deep neural networks to their respective inputs by computing attribution values for each dimension. In this way, we select the top 32 important dimensions for each medical image modality for late fusion and the final classifier for prediction. We evaluate our framework in BCI, IHC4BC-ER, and IHC4BC-PR datasets for cancer grading tasks.

Our contribution is the following:

- A framework with a virtual staining technique is designed to generate one more modality to enhance the limited, approximately paired input dataset without requiring pixel-level data alignment.

- We use a Shapley-value-based mechanism to reduce the dimensions of representation for enhanced multimodal fusion, thereby avoiding the curse of dimensionality and enhancing the interpretability of our multimodal technique.
- The experiment demonstrates that using virtual staining to generate an additional modality, combined with a Shapley-value-based dimension reduction technique, improves model performance. Specifically, it results in a 5% increase in accuracy for BCI, an 8% increase for IHC4BC-ER, and an 11% increase for IHC4BC-PR.

2 RELATED WORK

Previous general believes on H&E and IHC dataset.

Previous research primarily focuses on image translation and WSI registration algorithms, emphasizing the importance of precise pixel-level alignment for paired medical images (Liu et al., 2022). Competitions like ACROBAT (Weitz et al., 2023) have been organized to advance these technologies, particularly aligning H&E WSIs with IHC WSIs from identical tumor samples. Other studies (Naik et al., 2020; Anand et al., 2021; Shovon et al., 2022) suggest bypassing hard-to-obtain IHC images and predicting cancer and molecular biomarkers using only H&E whole slide images due to accessibility issues.

Virtual staining technique in medical images. Deep learning-based virtual staining technique has emerged as an exciting new field to provide more cost-effective, rapid, and sustainable solutions to histopathological tasks. However, there is no superior measurement standard in this field currently (Latonen et al., 2024). Many studies (Ozyoruk et al., 2022; Levy et al., 2021) rely on pathologists to manually assess the quality of virtually stained images. Others evaluate generated images using traditional metrics like PSNR, SSIM, and FID (de Haan et al., 2021; Vasiljević et al., 2022).

Multi-modal fusion in histopathology. Several studies (Chen et al., 2020; Li et al., 2022; Wang et al., 2021; Chen et al., 2022) have utilized multimodal techniques to combine histology and genomic data. More and more work design a very complex multimodal fusion framework (Wang et al., 2021; Huang et al., 2020; Lipkova et al., 2022; Stahlschmidt et al., 2022). However, there is a lack of research on using common stains like H&E and IHC in multimodal approaches.

3 FRAMEWORK DESIGN

Our whole framework consists of Parallel Feature Extraction Pipelines for each modality and a SHAP-CAT pipeline for the predictions of multimodal representations, as illustrated in Fig. 1. Given approximate H&E-IHC paired dataset I_{he}, I_{ihc} , we firstly use pre-trained CycleGAN to generate reconstructed H&E images $I_{rec.he}$. Then we separately train each modality to extract bag-level representations for each modality for further late fusion.

Modality Generation. Given the paucity of medical data in general (Zitnik et al., 2019; Miotto et al., 2018), the use of synthetic data has become increasingly prevalent for the training, development, and augmentation of artificial intelligence models (Latonen et al., 2024). We first use the virtual staining technique to generate another modality image for enhancing multimodal framework performance from H&E and IHC paired images, denoted as reconstructed H&E.

The virtual staining technique we used in our paper is CycleGAN (Zhu et al., 2017), which is specifically designed for unpaired datasets. The input of our framework is H&E-IHC approximate paired datasets I_{he}, I_{ihc} with labels. Approximate paired here means that these two sets of images are not aligned pixel to pixel, whereas the same pair of images are offset by about 10% in the vertical and horizontal directions. There are two translators $G : I_{he} \rightarrow I_{ihc}$, and $F : I_{ihc} \rightarrow I_{he}$ (as shown in Fig 1.a). G and F are trained simultaneously to encourages $F(G(I_{he})) \approx I_{he}$ and $G(F(I_{ihc})) \approx I_{ihc}$. Also, there are two adversarial discriminators D_{he} and D_{ihc} , where D_{he} aims to discriminate between images I_{he} and translated images $F(I_{ihc})$. Similarly, D_{ihc} aims to distinguish between I_{ihc} and $G(I_{he})$. The final objective is:

$$G^*, F^* = \arg \min_{G, F} \max_{D_{he}, D_{ihc}} \mathcal{L}(G, F, D_{he}, D_{ihc}). \quad (1)$$

The new modality, reconstructed from real H&E-IHC approximate paired images, forms a clinically and biologically relevant pair. Both IHC and reconstructed H&E offer different perspectives of the original H&E slide.

Algorithm 1 The framework of SHAP-CAT

Start with: Approximate paired H&E-IHC staining image I_{he}, I_{ihc} with labels y

- 1: Pre-train a CycleGAN by approximate paired H&E and IHC dataset.
 - 2: Reconstruct $\{I_{rec.he}\}_{n=1}^{N_{all}}$ from $\{I_{he}, I_{ihc}\}_{n=1}^{N_{all}}$ by pre-trained CycleGAN.
 - 3: Preprocess the WSIs $\{I_{he}, I_{ihc}, I_{rec.he}\}_{n=1}^{N_{all}}$ and extract features $\{R_{he}, R_{ihc}, R_{rec.he}\}_{n=1}^{N_{all}}$.
 - 4: Data splitting of $\{D\}_{n=1}^{N_{all}} \rightarrow \{D_1\}_{n=1}^{N_{train}}, \{D_2\}_{n=1}^{N_{val}}, \{D_3\}_{n=1}^{N_{test}}$.
 - 5: **while (Parallel Feature Extraction Pipeline) do**
 - 6: **for** each modality **do**
 - 7: $Model.fit(R, y)$ on $\{D_1\}_{n=1}^{N_{train}}$ with $\{D_2\}_{n=1}^{N_{val}}$
 - 8: $\hat{y} \leftarrow Model(R)$ on $\{D_3\}_{n=1}^{N_{test}}$ to obtain the performance for single modality pipeline
 - 9: extract bag-level representation z at the penultimate hidden layer
 - 10: **end for**
 - 11: **end while**
 - 12: **while (SHAP-CAT multimodal pipeline) do**
 - 13: Apply SHAP pooling σ to reduce dimensions for $z_{he}, z_{ihc}, z_{rec.he}$, respectively
 $f_{he} \leftarrow \sigma_{he}(z_{he}), f_{ihc} \leftarrow \sigma_{ihc}(z_{ihc}), f_{rec.he} \leftarrow \sigma_{rec.he}(z_{rec.he})$, where $f \in \mathbb{R}^{1 \times 32}$
 - 14: Concat two H&E representations: $f_{he.final} \leftarrow [f_{he}, f_{rec.he}]$, where $f_{he.final} \in \mathbb{R}^{1 \times 64}$
 - 15: Fusion three modalities: $F = f_{he.final} \otimes f_{ihc}$, where $F \in \mathbb{R}^{1 \times 2048}$
 - 16: **end while**
 - 17: Mapping of $F \rightarrow y$
 $y \leftarrow classifier(F)$ on $\{D_1\}_{n=1}^{N_{train}}$
 - 18: Obtaining the performance for SHAP-CAT multi-modality pipeline:
 $\hat{y} \leftarrow classifier(F)$ on $\{D_3\}_{n=1}^{N_{test}}$
-

Parallel Feature Extraction Pipeline. In this paper, the three modalities used—H&E, IHC, and reconstructed H&E images—are each assigned to a specific feature extraction pipeline. For each input WSI denoted as “bag” in the standard attention-based MIL pipeline (Ilse et al., 2018; Lu et al., 2021), the bag I is split into K patches $I = \{I(1), I(2), \dots, I(K)\}$, where x is denoted as “instance” and K varies for different input. Each bag will be pre-processed and then extract feature $R = \{r_1, r_2, \dots, r_K\}$. There are N such bag with their label y constituting the dataset $\{D\}_{n=1}^{N_{all}}$. During training, the whole dataset will be splitted into training $\{D_1\}_{n=1}^{N_{train}}$, validating $\{D_2\}_{n=1}^{N_{val}}$, and testing $\{D_3\}_{n=1}^{N_{test}}$ subset, where $\{I_{he}, I_{ihc}, I_{rec.he}\}_{n=1}^{N_{all}}$ sharing the same data splitting subset.

The embedding r_k is compressed by a fully-connected layer to h_k . Then h_k is fed into the multi-class classification network, aggregating the set of embeddings h_k into a bag-level embedding $z_n = \sum_{k=1}^K a_{k,n} h_k$, where the attention scores for the k -th instance is computed by Eq. 2.

$$a_{k,n} = \frac{\exp \{W_{atten,n}(\tanh(Vh_k^T) \odot \text{sigm}(Uh_k^T))\}}{\sum_{j=1}^K \exp \{W_{atten,n}(\tanh(Vh_j^T) \odot \text{sigm}(Uh_j^T))\}} \quad (2)$$

Finally, the bag-level representation z_n is extracted at the penultimate hidden layer before the last classifier.

SHAP-CAT Fusion Module. Once the bag-level representations have been constructed from each modality, a SHAP-CAT fusion module is introduced to capture informative inter-modality interactions between H&E, IHC, and reconstructed H&E features.

Before late fusion, we propose an efficient and highly interpretable SHAP pool to reduce dimensions of bag-level representations z to avoid the curse of dimensions. We model the dimension reduction as an attribution problem that attributes the prediction of machine learning models to their inputs (Lundberg & Lee, 2017; Ribeiro et al., 2016; Shrikumar et al., 2017). For bag-level representations $z = [d_1, d_2, \dots, d_{512}] \in \mathbb{R}^{1 \times 512}$, each dimension d has attribution values corresponding to the contributions toward the model prediction. Dimensions that have no effect on the output are assigned zero attribution, suggesting no relevance, whereas dimensions that significantly influence the output exhibit higher attribution values, indicating their importance. As illustrated in Fig 1(c), we visualize the attribution values of each dimension to understand the magnitude of how much it impacts the output.

The proposed SHAP pool selects the top 32 important dimensions for each modality and then applies the Kronecker product as late fusion. This module constructs the joint representations as the input of the final prediction for multimodalities. The whole algorithm is shown in Algorithm 1. We further introduce Shapley-value-based dimension reduction and multimodal fusion deeply in the next section.

4 EXPLAINABLE MULTI-MODAL FUSION

In this section, we define the impact of dimension reduction in multimodal technique as an attribution problem, quantifying how the changes of dimensions within input representations affect the model output.

4.1 PROBLEM FORMULATION

Given a set of inputs $\{z_n\}_{n=1}^N$ where $z = [d_1, d_2, \dots, d_{512}] \in \mathbb{R}^{1 \times 512}$ and a model $f(z)$, the output changes when dimensions within z vary. Each dimension d_i can interact with each other. Therefore, we define the attribution problem as follows: each dimension d_i has its attribution value ϕ_i , which indicates how much it impacts the output. The goal is to determine the attribution values $\{\phi_1, \phi_2, \dots, \phi_{512}\}$ of input bag-level representations $\{z_n\}_{n=1}^N$ by computing the contribution of each dimension within z to the model output. We simplify the problem into:

$$\{z_n\}_{n=1}^N = \begin{bmatrix} d_{1,1} & d_{1,2} & \cdots & d_{1,512} \\ \vdots & \vdots & \vdots & \vdots \\ d_{N,1} & d_{N,2} & \cdots & d_{N,512} \end{bmatrix} = \{x_1, x_2, \dots, x_{512}\} \Rightarrow \{\phi_1, \phi_2, \dots, \phi_{512}\} \quad (3)$$

In our paper, Shapley value (Shapley et al., 1953), which is a solution in game theory to denote a player’s marginal contribution to the payoff of a coalition game, is employed to measure the impact of individual dimension within representations for a model.

There is a characteristic function v that maps subsets $S \subseteq \{x_1, x_2, \dots, x_{512}\}$ to a real value $v(S)$, which represents how much payoff a set of dimensions can gain by ”comperating” as a set. $v(S)$ measures the importance of dimensions by sets. Now, we move on to the single dimension. The marginal contribution $\Delta_v(i, S)$ of the specific dimension features x_i with respect to a subset S is denoted as $\Delta_v(i, S) = v_{S \cup \{i\}}(x_{S \cup \{i\}}) - v_S(x_S)$. Intuitively, the Shapley value can be defined as the weighted average of the specific dimension’s marginal contributions to all possible subsets of dimensions.

Definition 1 *Shapley values quantifies the importance of each useful dimension by marginal contribution*

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v_{S \cup \{i\}}(x_{S \cup \{i\}}) - v_S(x_S)] \quad (4)$$

The above formula is a summation over all possible subsets S of feature values excluding the x_i ’s value. ϕ_i is a unique allocation of the coalition and can be viewed as the influence of x_i on the outcome. Therefore, the question becomes – how to identify $\{\phi_1, \phi_2, \dots, \phi_{512}\}$ of bag-level representations for a machine learning model.

4.2 INTERPRETABILITY IN MACHINE LEARNING

To obtain attribution values for each dimension, we need to explain the machine learning model first. For complex models in machine learning, its explanation can be represented by a simpler explanation model (Ribeiro et al., 2016; Lundberg & Lee, 2017; Shrikumar et al., 2017).

The simplified explanatory model is defined as an interpretable approximation of the original model. The original model that needs to be explained is given as f . g is the explanation model to explain f based on the single dimension x_i of feature: $f(x) = g(x')$. Explanation models distinguish an interpretable representation from the original feature space that the model uses. The function $x = h_x(x')$ is applied to map the original value x to a simplified input x' , where $x' \in \{0, 1\}^M$, M is the max number of coalition, and $\phi_i \in R$.

The simplified input x' maps 0 or 1 to the corresponding feature value, indicating the present or absent state of the corresponding feature value.

Definition 2 *Mapping feature value into simplified input*

$$x' = \left\{ (x^p)' = 1, \quad (x^a)' = 0 \begin{cases} x_i = 0 \\ x_i \neq 0, \text{ but } v_{S \cup \{i\}}(x_{S \cup \{i\}}) = v_S(x_S) \forall S \end{cases} \right\}$$

where x^p means the presence of a feature and x^a means the absence of a feature; we will discuss them in Section 4.3. ϕ_i is the attribution value of x_i , corresponding the the specific dimension d_i for bag-level representations. The function $x = h_x(x')$ maps 1 to the specific dimension that we want to explain and maps 0 to the values of the specific dimension that has no attributed impact on the model.

Property 1 *Meaningless dimension*

$$x'_i = 0 \Rightarrow \phi_i = 0 \quad (5)$$

After turning a feature vector into a discrete binary vector, we can define the attribution values for the model. For an explanatory model to have additive feature attribution, the explanatory model could be expressed as the sum of the null output of the model and the summation of explained effect attribution.

Property 2 *Local accuracy*

$$f(x) = g(x') = \phi_0 + \sum_{i=1}^M \phi_i x'_i \quad (6)$$

Explanation models also exhibit a property known as consistency, stating that if a model changes and makes a contribution of a particular feature stays the same or increases regardless of other inputs, the attribution assigned to that feature should not decrease.

Property 3 *Consistency*

$$v_1(S \cup \{i\}) - v_1(S) \geq v_2(S \cup \{i\}) - v_2(S) \forall S \Rightarrow \phi_i(v_1, x) \geq \phi_i(v_2, x) \quad (7)$$

Combining the information from Section 4.1 and 4.2, we can find that Shapley value is the only solution to satisfy the three properties of the explanatory models. Now, we get the explanation models related to attribution values ϕ_i . The new question is – how to estimate it?

4.3 SHAPLEY VALUE OF FEATURE DIMENSION

From all the previous property and definition, we have $\phi_0 = E[f(x)] = f(\emptyset)$. So the Property 2 will be $f(x) = g(x') = \sum_{i=1}^M \phi_i x'_i$, stating that when approximating the model f for input x , the

explanation’s attribution values ϕ_i for each feature x_i should sum up to the output $f(x)$. We aim to obtain local feature attributions ϕ_i , a vector of importance values for each feature of a model prediction for a specific sample x_i .

According to Definition 2, if feature x_i is present, we can simply set that feature to its value in x^p . The next step is to address the absence of a feature x^a .

One approach to incorporate x^a into the coalitional game is with a conditional expectation. We condition the set of features that are “present” as if we know them and use those to guess at the “missing” features, so the value of the game is: $v(S) = E_D[f(x)|x_S]$. Therefore,

$$\phi_i(f, x^p) = \frac{1}{|D|} \sum_{x^a \in D} \phi_i(f, x^p, x^a), \text{ where } D \text{ is the distribution of } x^a.$$

In summary, obtaining $\phi_i(f, x^p)$ reduces to an average of simpler problems $\phi_i(f, x^p, x^a)$, where our x^p is compared to a distribution with only one sample x^a .

In our paper, we employ treeSHAP (Lundberg et al., 2020), designed as a fast alternative for tree-based ML models such as random forests or decision trees, to calculate $\phi_i(f, x^p, x^a)$. Computational complexity is reduced to $O(TLD^2)$ where D is the maximum depth of any tree, L is the number of leaves and T is the number of trees.

Given bag-level representation $\{z_n\}_{n=1}^N \in \mathbb{R}^{N \times 512}$ with labels y , we train a random forest classifier on $\{z_n, y_n\}_{n=1}^N$ for estimation to obtain the attribution value $[\phi_1, \phi_2, \dots, \phi_{512}]$ for dimension reduction. The whole SHAP pool is demonstrated in Algorithm 2.

Algorithm 2 (SHAP pool)

Input: $\{z\}_{n=1}^{N_{all}}$ with label $\{y\}_{n=1}^{N_{all}}$, where $z = [d_1, d_2, \dots, d_{512}] \in \mathbb{R}^{1 \times 512}$

Output: $\{f\}_{n=1}^{N_{all}}$, where $f \in \mathbb{R}^{1 \times 32}$

- 1: $z_{train}, z_{test}, y_{train}, y_{test} \leftarrow Data_Split(z, y)$
 - 2: $model = RandomForestClassifier()$
 - 3: $model.fit(z_{train}, y_{train})$
 - 4: $shap_values \leftarrow treeSHAP(model, z_{test})$
 $[\phi_1, \phi_2, \dots, \phi_{512}] \leftarrow [x_1, x_2, \dots, x_{512}]$, where ϕ is the attribution value of each dimension
 - 5: select top 32 shap_values ϕ_i for z
 - 6: Dimension reduction:
 $f \leftarrow \sigma(z)$, where $z \in \mathbb{R}^{1 \times 512}$ and $f \in \mathbb{R}^{1 \times 32}$
-

4.4 FUSION OF MODALITY

In multi-modal fusion, direct fusion of multiple modalities is impractical. For example, bag-level representation of each modality is represented in 512 dimensions in our paper. Consequently, three dimensions would generate features of 512^3 dimensions, making it impractical for machine learning model training. In addition, such large-dimension data face a challenge known as the curse of dimensionality. Furthermore, trying to tackle complex histopathological tasks with such high-dimensional yet low-sample-size features results in “blind spot” (Berisha et al., 2021).

Therefore, we must decrease the dimensionality of representations. Prior research has utilized average pooling or max pooling for this purpose (Wang et al., 2024; Chen et al., 2020). Our method deviates from traditional methods by offering a more accurate and interpretable strategy for fusion. We are the first to implement a Shapley-value-based technique to reduce dimensions in image modality representations. We also evaluate our SHAP pool in a single modality by reducing bag-level representation $z \in \mathbb{R}^{1 \times 512}$ to $f \in \mathbb{R}^{1 \times 32}$ and then aggregated by different classifiers (as shown in Tab 1). We compare our SHAP pooling with average pooling, max pooling and selecting 32 dimensions randomly. Our SHAP pooling performs well across different classifiers.

Generate low-dimension features by SHAP Pooling. From the Parallel Feature Extraction Pipeline, we extract bag-level feature representations z_{he} , z_{rec_he} and z_{ihc} .

We adopt the proposed shaply-valuse-based pooling to fuse H&E, IHC and reconstructed H&E representations.

Table 1: Effectiveness of Proposed Shap Pooling.

Model	Accuracy					
	SHAP	Avg	Max	Rand1	Rand2	Rand3
Random Forest	0.898	0.867	0.849	0.821	0.829	0.808
SVM	0.900	0.862	0.852	0.785	0.795	0.762
Logistic Regression	0.862	0.765	0.793	0.734	0.698	0.734
KNeighbors	0.903	0.903	0.893	0.793	0.847	0.806
Decision Tree	0.824	0.760	0.777	0.734	0.739	0.721
MLP (ours)	0.885	0.821	0.859	0.777	0.806	0.767
XGB Classifier	0.903	0.882	0.875	0.816	0.847	0.811
LGBM Classifier	0.900	0.885	0.880	0.818	0.849	0.813
CatBoost (ours)	0.903	0.875	0.880	0.839	0.847	0.844

Using SHAP pooling σ , we select the most important 32 dimensions of original bag-level representation z_{he} , $z_{rec.he}$ and z_{ihc} to generate low-dimension representation f_{he} , $f_{rec.he}$ and $f_{ihc} \in \mathbb{R}^{1 \times 32}$ by $f = \sigma(z)$.

Kronecker product. IHC is a staining technique that visualizes the overexpression of target proteins. The visualized locations help understand the morphological characteristics of cells within a tissue. Thus, IHC and H&E WSIs provide different information on molecular features. For the modality from true H&E and reconstructed H&E whole slide images, z_{he} and $z_{rec.he}$ are directly concatenated to generate the new representation $f_{he.final} \in \mathbb{R}^{1 \times 64}$ of H&E staining images: $f_{he.final} = [f_{he}, f_{rec.he}] = [\sigma(z_{he}), \sigma(z_{rec.he})]$. In order to capture the intricate relationships between H&E and IHC modalities, we follow previous work (Wang et al., 2021; Chen et al., 2020; Wang et al., 2024; Chen et al., 2022; Li et al., 2022) that employ Kronecker product, denoted as $\otimes$, to fuse different modalities.

Therefore, the joint multimodal tensor $F \in \mathbb{R}^{1 \times 2048}$ constructed from the Kronecker product, as shown in Eq.(8), will capture the important interactions that characterize H&E and IHC modalities.

$$F = f_{he.final,n} \otimes f_{ihc,n} = [\sigma(z_{he}), \sigma(z_{rec.he})] \otimes \sigma(z_{ihc,n}) \quad (8)$$

After constructing the joint representation, we use the multimodal representation F as input. It is then processed by classifiers like MLP or CatBoost (Prokhorenkova et al., 2018) for cancer grading tasks.

5 EXPERIMENTS

Datasets and Implementation Details We use two public datasets BCI (Liu et al., 2022) and IHC4BC (Akbarnejad et al., 2023) in this paper. Both of them are cancer grading tasks. We use CLAM (Lu et al., 2021) as the pre-processing tool and original training pipeline. The details are shown in the Appendix.

Results on BCI and IHC4BC datasets Tab 2 shows the detailed results on the BCI dataset, and Tab 3 presents the results on the IHC4BC-ER and IHC4BC-PR datasets. Most previous models only deal with a single modality. Multiple modalities achieve higher performance than all models in a single modality. Our SHAP-CAT method includes modality enhancement via virtual staining, efficient multimodal fusion by Shapley-value-based dimension reduction, and finally, aggregation in the MLP or CatBoost classifiers (Prokhorenkova et al., 2018), achieving higher accuracy across BCI and IHC4BC datasets.

Reconstruct modality enhance the performance for multimodal model As mentioned in the previous section, our framework uses the CLAM pipeline to extract the bag-level representations z . Therefore, we also report the performance of the baseline trained by reconstructed H&E modality. As shown in Tab 2 and Tab 3, the reconstructed H&E modality generated by CycleGAN results in lower performance when it is used as the main input for the single-modality model. However, it can enhance multimodal model performance when we use our SHAP-CAT fusion to efficiently

Table 2: **Experiment Results on the BCI Dataset.** The performance is reported as AUC and ACC.

Model	Modality	Performance	
		AUC	ACC
InceptionV3 (Szegedy et al., 2016)	H&E	0.823	0.804
ResNet (He et al., 2016)	H&E	0.886	0.872
ViT (Ayana et al., 2023)	H&E	0.92	0.904
HAHNet (Wang et al., 2023)	H&E	0.99	0.937
DenseNet (Huang et al., 2017)	H&E	0.890	0.68
HE-HER2Net (Shovon et al., 2022)	H&E	0.980	0.870
ABMIL (Ilse et al., 2018)	H&E	0.985	0.902
ABMIL	IHC	0.991	0.916
CLAM (Lu et al., 2021)	H&E	0.987	0.909
CLAM	IHC	0.991	0.917
TransMIL (Shao et al., 2021)	H&E	0.991	0.907
TransMIL	IHC	0.994	0.931
Shap-cat Fusion + MLP (ours)	H&E, rec H&E, IHC	0.997	0.959
Shap-cat Fusion + CatBoost (ours)	H&E, rec H&E, IHC	0.996	0.955

capture information across three modalities. In the BCI dataset, three original pipelines, which train H&E, IHC, and rec H&E modalities separately to extract bag-level representations $z_{he}, z_{ihc}, z_{rec.he}$, achieve accuracy in 0.909, 0.917 and 0.787. However, their multimodal representations can be aggregated by the classifier in much higher results, achieving 0.959 in accuracy. This situation also occurs in IHC4BC datasets.

Table 3: **Experiment Results on IHC4BC Dataset.** The performance is reported as AUC and ACC for IHC4BC-ER and IHC4BC-PR.

Model	Modality	IHC4BC-ER		IHC4BC-PR	
		AUC	ACC	AUC	ACC
ABMIL	H&E	0.953	0.843	0.911	0.835
ABMIL	IHC	0.978	0.888	0.959	0.841
CLAM	H&E	0.9543	0.8421	0.908	0.777
CLAM	IHC	0.979	0.894	0.957	0.84
Transmil	H&E	0.95	0.851	0.911	0.791
Transmil	IHC	0.979	0.902	0.959	0.85
Shap-cat Fusion + MLP (ours)	H&E, rec H&E, IHC	0.98	0.925	0.921	0.877
Shap-cat Fusion + CatBoost (ours)	H&E, rec H&E, IHC	0.985	0.928	0.969	0.883

6 ABLATION STUDY

In our paper, we use the following strategies:

- Strategy 1 : virtual staining to generate reconstructed H&E
- Strategy 2 : SHAP pooling to reduce the dimension of original bag-level representation

We evaluate our virtual staining strategy. Since we use CLAM to extract bag-level representations, we compare single, double, and triple modalities in Table 4. Also, we compare the results of two modalities(H&E-IHC) with three modalities(H&E, IHC, and reconstructed H&E) processed by the same pooling across different classifiers as aggregations in Table 5. What’s more, we evaluate our SHAP pool with average pool across different classifiers in Table 5.

Table 4: **Ablation Study of Virtual Staining on BCI and IHC4BC datasets.** Results are reported as AUC and ACC for each modality.

Model	Modality	BCI		IHC4BC-ER		IHC4BC-PR	
		AUC	ACC	AUC	ACC	AUC	ACC
CLAM	H&E	0.987	0.909	0.954	0.842	0.908	0.777
CLAM	IHC	0.991	0.917	0.979	0.894	0.957	0.84
CLAM	rec H&E	0.937	0.787	0.949	0.835	0.916	0.783
Shap-cat + MLP	H&E, IHC	0.995	0.941	0.984	0.91	0.919	0.866
Shap-cat + CatBoost	H&E, IHC	0.994	0.946	0.985	0.911	0.967	0.875
Shap-cat + MLP	H&E, rec H&E, IHC	0.997	0.959	0.98	0.925	0.921	0.877
Shap-cat + CatBoost	H&E, rec H&E, IHC	0.996	0.955	0.985	0.928	0.969	0.883

Table 5: **Ablation Study of SHAP pooling on BCI dataset.** The results are reported as the average AUC and ACC metrics for two and three multimodal settings.

Framework	Model	Two Multimodal		Three Multimodal	
		AUC	ACC	AUC	ACC
Avg Bilinear Fusion	RandomForest	0.994	0.936	0.997	0.941
	Logistic Regression	0.982	0.844	0.986	0.890
	DecisionTree	0.898	0.857	0.892	0.844
	MLP	0.989	0.923	0.995	0.944
	GaussianNB	0.963	0.872	0.949	0.862
	CatBoostClassifier	0.990	0.928	0.996	0.944
Shap-cat Multimodal Fusion	RandomForest	0.993	0.944	0.996	0.951
	Logistic Regression	0.995	0.928	0.994	0.932
	DecisionTree	0.891	0.872	0.903	0.883
	MLP	0.995	0.941	0.997	0.959
	GaussianNB	0.967	0.918	0.996	0.956
	CatBoostClassifier	0.994	0.946	0.996	0.955

7 DISCUSSION

7.1 VIRTUAL STAINING CAN BE USED FOR ENHANCING NOT MAIN INPUT

Limited labeled datasets are a crucial challenge for the whole histopathology field, especially for multimodal models. Our framework applies a virtual staining technique to enhance WSI classification, providing a different solution. Our synthesis data satisfy the following requirements suggested by the FDA AI/ML white paper and 21st Century Cures Act (Steyaert et al., 2023): (1) relevant to the clinical practice and clinical endpoint; (2) collected in a manner that is consistent, generalizable, and clinically relevant; and (3) output is appropriately transparent for users.

We claim that virtual staining may not be good for the training model as the main input, but it is good for enhancing performance as an extra modality. As shown in Table 5, our reconstructed modality performs well across different classifiers, compared to the single or double modality.

7.2 DENSE BAG-LEVEL REPRESENTATION

SHAP value (Lundberg & Lee, 2017) is an approximation of Shapley values, while the original Shapley value (Shapley et al., 1953) is an NP-hard problem in game theory. It is impossible to search for an NP-hard problem directly in features extracted from giga-pixel WSIs. The bag-level representation generated by our framework is a 512-dimension feature with a size of 3.3kb. There are 1×512 elements in the tensor. Each element is 4 bytes. Therefore, the total data size is 2048 bytes (or exactly 2 KB for the data). The raw data size is about 2 KB. The actual file size might be slightly larger due to the metadata and can vary slightly based on the specific version of PyTorch and the details of how the tensor storage is implemented. Similarly, our final bag-level representation is a 2048-dimension tensor, which is only 9.4kb in size. This very small size ensures us to use many models to aggregate the final bag-level representation.

8 CONCLUSION

We propose a novel framework with a virtual staining technique to generate one more modality to enhance WSI classification and a Shapley-value-based mechanism to reduce dimensions for efficient and interpretable multimodal fusion for histopathological tasks. We are the first to use the Shapley-value-based dimension-reducing technique in image modality. The experiment demonstrates that using virtual staining to generate an additional modality, combined with a Shapley-value-based dimension reduction technique, improves model performance. Specifically, it results in a 5% increase in accuracy for BCI, 8% increase for IHC4BC-ER and a 11% increase for IHC4BC-PR.

REFERENCES

- Amir Akbarnejad, Nilanjan Ray, Penny J Barnes, and Gilbert Bigras. Predicting ki67, er, pr, and her2 statuses from h&e-stained breast cancer images. *arXiv preprint arXiv:2308.01982*, 2023.
- Hani A Alturkistani, Faris M Tashkandi, and Zuhair M Mohammedsaleh. Histological stains: a literature review and case study. *Global journal of health science*, 8(3):72, 2016.
- Deepak Anand, Kumar Yashashwi, Neeraj Kumar, Swapnil Rane, Peter H Gann, and Amit Sethi. Weakly supervised learning on unannotated h&e-stained slides predicts braf mutation in thyroid cancer with high accuracy. *The Journal of pathology*, 255(3):232–242, 2021.
- Gelan Ayana, Eonjin Lee, and Se-woon Choe. Vision transformers for breast cancer human epidermal growth factor receptor 2 expression staging without immunohistochemical staining. *The American Journal of Pathology*, 2023.
- Visar Berisha, Chelsea Krantsevich, P Richard Hahn, Shira Hahn, Gautam Dasarathy, Pavan Turaga, and Julie Liss. Digital medicine and the curse of dimensionality. *NPJ digital medicine*, 4(1):153, 2021.
- Kevin M Boehm, Pegah Khosravi, Rami Vanguri, Jianjiong Gao, and Sohrab P Shah. Harnessing multimodal data integration to advance precision oncology. *Nature Reviews Cancer*, 22(2):114–126, 2022.
- Richard J Chen, Ming Y Lu, Jingwen Wang, Drew FK Williamson, Scott J Rodig, Neal I Lindeman, and Faisal Mahmood. Pathomic fusion: an integrated framework for fusing histopathology and genomic features for cancer diagnosis and prognosis. *IEEE Transactions on Medical Imaging*, 41(4):757–770, 2020.
- Richard J Chen, Ming Y Lu, Drew FK Williamson, Tiffany Y Chen, Jana Lipkova, Zahra Noor, Muhammad Shaban, Maha Shady, Mane Williams, Bumjin Joo, et al. Pan-cancer integrative histology-genomic analysis via multimodal deep learning. *Cancer Cell*, 40(8):865–878, 2022.
- Kevin de Haan, Yijie Zhang, Jonathan E Zuckerman, Tairan Liu, Anthony E Sisk, Miguel FP Diaz, Kuang-Yu Jen, Alexander Nobori, Sofia Liou, Sarah Zhang, et al. Deep learning-based transformation of h&e stained tissues into special stains. *Nature communications*, 12(1):1–13, 2021.
- Chaitanya Dwivedi, Shima Nofallah, Maryam Pouryahya, Janani Iyer, Kenneth Leidal, Chuhan Chung, Timothy Watkins, Andrew Billin, Robert Myers, John Abel, et al. Multi stain graph fusion for multimodal integration in pathology. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 1835–1845, 2022.
- Sebastian Foersch, Christina Glasner, Ann-Christin Woerl, Markus Eckstein, Daniel-Christoph Wagner, Stefan Schulz, Franziska Kellers, Aurélie Fernandez, Konstantina Tserea, Michael Kloth, et al. Multistain deep learning for prediction of prognosis and therapy response in colorectal cancer. *Nature medicine*, 29(2):430–439, 2023.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.

-
- Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger. Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 4700–4708, 2017.
- Shih-Cheng Huang, Anuj Pareek, Saeed Seyyedi, Imon Banerjee, and Matthew P Lungren. Fusion of medical imaging and electronic health records using deep learning: a systematic review and implementation guidelines. *NPJ digital medicine*, 3(1):136, 2020.
- Maximilian Ilse, Jakub Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *International conference on machine learning*, pp. 2127–2136. PMLR, 2018.
- Guillaume Jaume, Anurag Vaidya, Andrew Zhang, Andrew H Song, Richard J Chen, Sharifa Sahai, Dandan Mo, Emilio Madrigal, Long Phi Le, and Faisal Mahmood. Multistain pretraining for slide representation learning in pathology. *arXiv preprint arXiv:2408.02859*, 2024.
- Leena Latonen, Sonja Koivukoski, Umair Khan, and Pekka Ruusuvuori. Virtual staining for histology by deep learning. *Trends in Biotechnology*, 2024.
- Joshua J Levy, Nasim Azizgolshani, Michael J Andersen Jr, Arief Suriawinata, Xiaoying Liu, Mikhail Lisovsky, Bing Ren, Carly A Bobak, Brock C Christensen, and Louis J Vaickus. A large-scale internal validation study of unsupervised virtual trichrome staining technologies on nonalcoholic steatohepatitis liver biopsies. *Modern Pathology*, 34(4):808–822, 2021.
- Ruiqing Li, Xingqi Wu, Ao Li, and Minghui Wang. Hfbsurv: hierarchical multimodal fusion with factorized bilinear models for cancer survival prediction. *Bioinformatics*, 38(9):2587–2594, 2022.
- Jana Lipkova, Richard J Chen, Bowen Chen, Ming Y Lu, Matteo Barbieri, Daniel Shao, Anurag J Vaidya, Chengkuan Chen, Luoting Zhuang, Drew FK Williamson, et al. Artificial intelligence for multimodal data integration in oncology. *Cancer cell*, 40(10):1095–1110, 2022.
- Shengjie Liu, Chuang Zhu, Feng Xu, Xinyu Jia, Zhongyue Shi, and Mulan Jin. Bci: Breast cancer immunohistochemical image generation through pyramid pix2pix. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 1815–1824, June 2022.
- Ming Y Lu, Drew FK Williamson, Tiffany Y Chen, Richard J Chen, Matteo Barbieri, and Faisal Mahmood. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nature biomedical engineering*, 5(6):555–570, 2021.
- Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30, 2017.
- Scott M Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable ai for trees. *Nature machine intelligence*, 2(1):56–67, 2020.
- Riccardo Miotto, Fei Wang, Shuang Wang, Xiaoqian Jiang, and Joel T Dudley. Deep learning for healthcare: review, opportunities and challenges. *Briefings in bioinformatics*, 19(6):1236–1246, 2018.
- Nikhil Naik, Ali Madani, Andre Esteva, Nitish Shirish Keskar, Michael F Press, Daniel Ruderman, David B Agus, and Richard Socher. Deep learning-enabled breast cancer hormonal receptor status determination from base-level h&e stains. *Nature communications*, 11(1):1–8, 2020.
- Kutsev Bengisu Ozyoruk, Sermet Can, Berkan Darbaz, Kayhan Başak, Derya Demir, Guliz Irem Gokceler, Gurdeniz Serin, Uguray Payam Hacisalihoglu, Emirhan Kurtuluş, Ming Y Lu, et al. A deep-learning model for transforming the style of tissue images from cryosectioned to formalin-fixed and paraffin-embedded. *Nature Biomedical Engineering*, 6(12):1407–1419, 2022.
- Raquel Perez-Lopez, Narmin Ghaffari Laleh, Faisal Mahmood, and Jakob Nikolas Kather. A guide to artificial intelligence for cancer researchers. *Nature Reviews Cancer*, pp. 1–15, 2024.

-
- Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. Catboost: unbiased boosting with categorical features. *Advances in neural information processing systems*, 31, 2018.
- Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. ” why should i trust you?” explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, pp. 1135–1144, 2016.
- Zhuchen Shao, Hao Bian, Yang Chen, Yifeng Wang, Jian Zhang, Xiangyang Ji, et al. Transmil: Transformer based correlated multiple instance learning for whole slide image classification. *Advances in neural information processing systems*, 34:2136–2147, 2021.
- Lloyd S Shapley et al. A value for n-person games. 1953.
- Md Sakib Hossain Shovon, Md Jahidul Islam, Mohammed Nawshar Ali Khan Nabil, Md Mohimen Molla, Akinul Islam Jony, and MF Mridha. Strategies for enhancing the multi-stage classification performances of her2 breast cancer from hematoxylin and eosin images. *Diagnostics*, 12(11):2825, 2022.
- Avanti Shrikumar, Peyton Greenside, and Anshul Kundaje. Learning important features through propagating activation differences. In *International conference on machine learning*, pp. 3145–3153. PMIR, 2017.
- Sören Richard Stahlschmidt, Benjamin Ulfenborg, and Jane Synnergren. Multimodal deep learning for biomedical data fusion: a review. *Briefings in Bioinformatics*, 23(2):bbab569, 2022.
- Sandra Steyaert, Marija Pizurica, Divya Nagaraj, Priya Khandelwal, Tina Hernandez-Boussard, Andrew J Gentles, and Olivier Gevaert. Multimodal data fusion for cancer biomarker discovery with deep learning. *Nature machine intelligence*, 5(4):351–362, 2023.
- Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2818–2826, 2016.
- Jelica Vasiljević, Zeeshan Nisar, Friedrich Feuerhake, Cédric Wemmert, and Thomas Lampert. Cylegan for virtual stain transfer: Is seeing really believing? *Artificial Intelligence in Medicine*, 133:102420, 2022.
- Jiahao Wang, Xiaodong Zhu, Kai Chen, Lei Hao, and Yuaning Liu. Hahnet: a convolutional neural network for her2 status classification of breast cancer. *BMC bioinformatics*, 24(1):353, 2023.
- Jun Wang, Yu Mao, Yufei Cui, Nan Guan, and Chun Jason Xue. Ihc matters: Incorporating ihc analysis to h&e whole slide image analysis for improved cancer grading via two-stage multimodal bilinear pooling fusion. *arXiv preprint arXiv:2405.08197*, 2024.
- Zhiqin Wang, Ruiqing Li, Minghui Wang, and Ao Li. Gpbn: deep bilinear network integrating both genomic data and pathological images for breast cancer prognosis prediction. *Bioinformatics*, 37(18):2963–2970, 2021.
- Philippe Weitz, Masi Valkonen, Leslie Solorzano, Circe Carr, Kimmo Kartasalo, Constance Boissin, Sonja Koivukoski, Aino Kuusela, Dusan Rasic, Yanbo Feng, et al. A multi-stain breast cancer histological whole-slide-image data set from routine diagnostics. *Scientific Data*, 10(1):562, 2023.
- Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks. In *Proceedings of the IEEE international conference on computer vision*, pp. 2223–2232, 2017.
- Marinka Zitnik, Francis Nguyen, Bo Wang, Jure Leskovec, Anna Goldenberg, and Michael M Hoffman. Machine learning for integrating data in biology and medicine: Principles, practice, and opportunities. *Information Fusion*, 50:71–91, 2019.

A APPENDIX

A.1 DATASET

We use two public breast cancer datasets in this paper. BCI dataset (Liu et al., 2022) presents 4870 registered H&E and IHC pairs, covering a variety of HER2 expression levels from 0 to 3. IHC4BC dataset (Akbarnejad et al., 2023) contains H&E and IHC pairs in ER and PR breast cancer assessment, and categories are defined ranges 0 to 3 respectively. The number of each subset is 26135 and 24972.

A.2 IMPLEMENTATION DETAILS

We use CLAM (Lu et al., 2021) pre-processing tools to create patches and extract features from each WSI image. Some WSIs will be dropped due to the segment and filtering of the CLAM pre-processing mechanism; we take the intersection of H&E and IHC pre-processed WSIs for further training. The learning rate of the Adam optimizer is set to 2×10^{-4} , the weight decay is set to 1×10^{-5} , the early-stop strategy is used, and the max training epochs are 200. We trained our multimodal model using a weakly supervised paradigm in 5-fold Monte Carlo cross-validation and performed ablation analysis to compare the performance between unimodal and multimodal prognostic models. For each cross-validated fold, we randomly split each dataset into 80%-10%-10% subset of training, validation, and testing, stratified by each class.